

Foundation Programme
Universiti Teknologi Malaysia
Session 2019/2020 Semester 2

Test 2 (20%)

Time: 90 minutes

FSPM 0024 Calculus

Answer all questions.

1. The derivative of a function $g(x)$ is given as

$$\frac{d(g(x))}{dx} = f(x) + \frac{3}{x^4} + \sin x.$$

- i. Express $\int f(x) dx$ in terms of $g(x)$ and x .

(2 marks)

- ii. Hence, if $g\left(\frac{\pi}{2}\right) = 0$ and $g(\pi) = 1$, find the value of $\int_{\frac{\pi}{2}}^{\pi} f(x) dx$.

(3 marks)

2. Using appropriate substitution, evaluate

$$\int \frac{6x^3}{\sqrt{x^2 + 2}} dx.$$

(6 marks)

3. Evaluate

(a) $\int x^2 \sin 4x dx.$

(5 marks)

(b) $\int_4^5 \frac{2x}{x^2 - 4x + 3} dx.$

(6 marks)

4. By using substitution $x = \tan \theta$, evaluate

$$\int_0^1 \frac{x^2}{(1 + x^2)^2} dx.$$

(7 marks)

$$1) \frac{d}{dx} g(x) = f(x) + \frac{3}{x^4} + \sin x$$

$$g(x) = \int f(x) + \int \frac{3}{x^4} + \int \sin x \, dx$$

$$\int f(x) \, dx = g(x) + \frac{1}{x^3} + \cos x + C$$

$$\int_{\frac{\pi}{2}}^{\pi} f(x) \, dx = \int_{\frac{\pi}{2}}^{\pi} g(x) - \int_{\frac{\pi}{2}}^{\pi} \frac{3}{x^4} - \int_{\frac{\pi}{2}}^{\pi} \sin x$$

$$= g(\pi) - g\left(\frac{\pi}{2}\right) + \left[\frac{1}{x^3}\right]_{\frac{\pi}{2}}^{\pi} + [\cos x]_{\frac{\pi}{2}}^{\pi}$$

$$= 1 + 0.23 + 1$$

$$= 2.23$$

$$2) u^2 = x^2 + 2, \quad x^2 = u^2 - 2$$

$$\frac{du}{dx} 2u = 2x$$

$$dx = \frac{2u}{2x} \cdot du$$

$$= \frac{u}{x} \cdot du$$

$$\int \frac{6x^3}{\sqrt{x^2+2}} \, dx = \int \frac{6x^5}{\sqrt{u^2}} \cdot \frac{1}{x} \, du = \int 6x^2 \, du = \int 6u^2 - 12 \, du$$

$$= 3u^3 - 12u + C$$

$$= 3(x^2+2)^{\frac{3}{2}} - 12\sqrt{x^2+2} + C$$

$$3) (a) \int x^2 \sin 4x \, dx = uv - \int v \frac{du}{dx} = -\frac{x^2 \cos 4x}{4} - \int \frac{-2x \cos 4x}{4} = -\frac{x^2 \cos 4x}{4} + \frac{1}{2} \int x \cos 4x$$

$$u = x^2 \quad \frac{du}{dx} = \sin 4x$$

$$\frac{du}{dx} = 2x \quad v = \frac{-\cos 4x}{4}$$

$$u = x \quad \frac{du}{dx} = \cos 4x$$

$$\frac{du}{dx} = 1 \quad v = \frac{\sin 4x}{4}$$

$$= -\frac{x^2 \cos 4x}{4} + \frac{x \sin 4x}{4} - \int \frac{\sin 4x}{4}$$

$$= -\frac{x^2 \cos 4x}{4} + \frac{x \sin 4x}{4} + \frac{\cos 4x}{16}$$

$$(b) \int_4^5 \frac{2x}{x^2-4x+3} \, dx = \int_4^5 \frac{3}{x-3} \, dx - \int_4^5 \frac{1}{x-1} \, dx = 3[\ln(x-3)]_4^5 - [\ln(x-1)]_4^5 = 3(0.693-0) -$$

$$\frac{2x}{(x-3)(x-1)} = \frac{A}{x-3} + \frac{B}{x-1} = \frac{3}{x-3} - \frac{1}{x-1} \quad (1.386 - 1.099)$$

$$A(x-1) + B(x-3) = 2x$$

$$= 2.079 - 0.287$$

$$(A+B)x + (-A-3B) = 2x$$

$$= 1.792$$

$$A+B = 2 \quad -2+B-3B = 0$$

$$A = 2-B \quad -2B = 2$$

$$A = 3$$

$$B = -1$$

$$4) x = \tan \theta, \quad \int \frac{x^2}{(1+x^2)^2} \, dx = \int \frac{\tan^2 \theta}{(1+\tan^2 \theta)^2} \cdot \sec^2 \theta \, d\theta$$

$$\frac{dx}{d\theta} = \sec^2 \theta \quad \frac{du}{d\theta} = -\sin \theta \quad = \int \frac{\tan^2 \theta}{(\sec^2 \theta)^2} \cdot \sec^2 \theta \, d\theta$$

$$dx = \sec^2 \theta \, d\theta \quad d\theta = \frac{du}{-\sin \theta} \quad = \int \frac{\tan^2 \theta}{\sec^2 \theta} \, d\theta$$

$$x=1, \quad \theta = 0.785$$

$$= \int \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \cos^2 \theta \, d\theta$$

$$x=0, \quad \theta = 0$$

$$= \int \sin^2 \theta \, d\theta$$

$$= \frac{1}{2} \int 1 - \cos 2\theta \, d\theta = \frac{1}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) + C$$

$$\int_0^1 \frac{x^2}{(1+x^2)^2} \, dx = \frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{0.785}$$

$$= \frac{1}{2} (0.285 + C)$$

$$= 0.1425$$

5. Find the area of the region bounded by the curve $y = x^3$, the lines $y = 1$ and $y = 8$ and the y -axis.

(5 marks)

6. Sketch the region bounded by the curve $y = x^2 + 1$, the x -axis and the ordinates $x = -1$ and $x = 1$. If the region is rotated through 2π radians about x -axis to form a solid of revolution. Calculate the volume of this solid.

(6 marks)

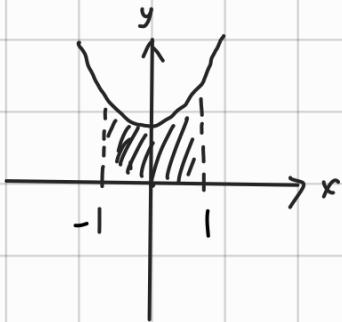
LIST OF FORMULA

Trigonometric	Differentiations	Integrations
$\cos^2 x + \sin^2 x = 1$ $1 + \tan^2 x = \sec^2 x$ $\cot^2 x + 1 = \operatorname{cosec}^2 x$ $\sin 2x = 2 \sin x \cos x$ $\cos 2x = \cos^2 x - \sin^2 x$ $\quad = 2 \cos^2 x - 1$ $\quad = 1 - 2 \sin^2 x$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$	$\frac{d}{dx}[k] = 0, k \text{ constant}$ $\frac{d}{dx}[x^n] = nx^{n-1}$ $\frac{d}{dx}[e^x] = e^x$ $\frac{d}{dx}[\ln x] = \frac{1}{x}$ $\frac{d}{dx}[\cos x] = -\sin x$ $\frac{d}{dx}[\sin x] = \cos x$ $\frac{d}{dx}[\tan x] = \sec^2 x$ $\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$ $\frac{d}{dx}[\sec x] = \sec x \tan x$ $\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$	$\int k dx = kx + C, k \text{ constant}$ $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$ $\int e^x dx = e^x + C$ $\int \frac{dx}{x} = \ln x + C$ $\int \sin x dx = -\cos x + C$ $\int \cos x dx = \sin x + C$ $\int \sec^2 x dx = \tan x + C$ $\int \operatorname{cosec}^2 x dx = -\cot x + C$ $\int \sec x \tan x dx = \sec x + C$ $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$
Logarithm		
$a^x = e^{x \ln a}$ $\log_a x = \frac{\log_b x}{\log_b a}$		

5) $y = x^3, y = 1, y = 8, x = \sqrt[3]{y}$

$$\text{Area} = \int_1^8 \sqrt[3]{y} = \int_1^8 y^{\frac{1}{3}} = \left[\frac{3}{4} y^{\frac{4}{3}} \right]_1^8 = 12 - \frac{3}{4} = 11.25 \text{ unit}^2$$

6)



$$\begin{aligned} \text{Volume} &= \left| \pi \int_0^1 (x^2 + 1)^2 dx \right| + \left| \pi \int_{-1}^0 (x^2 + 1)^2 dx \right| \\ &= \left| \pi \int_0^1 x^4 + 2x^2 + 1 dx \right| + \left| \pi \int_0^1 x^4 + 2x^2 + 1 dx \right| \\ &= \left| \pi \left[\frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^1 \right| + \left| \pi \left[\frac{x^5}{5} + \frac{2x^3}{3} + x \right]_{-1}^0 \right| \\ &= \left| \pi \left(\frac{28}{15} \right) \right| + \left| \pi \left(-\frac{28}{15} \right) \right| \\ &= 11.73 \text{ unit}^3 \end{aligned}$$